THE CLOSED SUBSCHEME CUT OUT BY THE ZEROTH FITTING IDEAL AGREES WITH THE SCHEME THEORETIC SUPPORT AND CARRIES THE MODULE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Statement span 1309-1326 and proof span 1328-1349 come from the halfA pool card operations/pool-cards/stacks-divisors-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof offers two ways of producing the module of part (3) with an 'either ... or ...' wording; both routes are the author's and are preserved.

Editorial note. 4 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The scheme theoretic support of a finite type quasi-coherent module and its local description by the annihilator, as defined and established in the morphisms chapter. Cited at proof lines 1329-1332 and 1338-1340. Identifies the scheme theoretic support and its support as a closed subset, which are two of the claims of the statement.
- The elementary properties of Fitting ideals, in particular that the zeroth Fitting ideal annihilates the module and that the closed subscheme it cuts out has the same underlying set as the support. Cited at proof lines 1332-1334 and 1340-1342. Gives the scheme theoretic inclusion of the support in the Fitting subscheme and the equality of the two underlying closed subsets.
- The equivalence between quasi-coherent modules on a closed subscheme and quasi-coherent modules annihilated by the ideal sheaf, cited from the morphisms chapter. Cited at proof lines 1344-1347. Produces the finite type module on the Fitting subscheme whose pushforward is the given module.

Source reference. Original lemma more-algebra-lemma-fitting-ideal-basics: sources/stacks-divisors/pinned-original-source.tex, source lines 1266–1266. Same-chapter reference used in the retained author text at source line(s) 1266, 1294, 1306, 1334, 1340, 1348. It is supplied only as a navigation pointer so that the retained text compiles; the

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Source reference. Original definition 07U1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9956–9956. Same-chapter reference used in the retained author text at source line(s) 1330. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 04CJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5693–5693. Same-chapter reference used in the retained author text at source line(s) 1345, 2728, 4308. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 07U0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9955–9955. Same-chapter reference used in the retained author text at source line(s) 1331, 1337, 1343. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let S be a scheme. Let $\mathcal{F}$ be a finite type, quasi-coherent $\mathcal{O}_S$-module. Let $Z_0 \subset S$ be the closed subscheme cut out by $Fit_0(\mathcal{F})$. Let $Z \subset S$ be the scheme theoretic support of $\mathcal{F}$. Then*

(1) *$Z \subset Z_0 \subset S$ as closed subschemes,*
(2) *$Z = Z_0 = Supp(\mathcal{F})$ as closed subsets,*
(3) *there exists a finite type, quasi-coherent $\mathcal{O}_{Z_0}$-module $\mathcal{G}_0$ with*

$$(Z_0 \to X)_*\mathcal{G}_0 = \mathcal{F}.$$

Proof. Recall that Z is locally cut out by the annihilator of $\mathcal{F}$, see Morphisms, Definition 07U1 (which uses Morphisms, Lemma 07U0 to define Z). Hence we see that $Z \subset Z_0$ scheme theoretically by More on Algebra, Lemma more-algebra-lemma-fitting-ideal-basics part (6). On the other hand we have $Z = \mathrm{Supp}(\mathcal{F})$ set theoretically by Morphisms, Lemma 07U0 and we have $Z_0 = Z$ set theoretically by More on Algebra, Lemma more-algebra-lemma-fitting-ideal-basics part (7). Finally, to get $\mathcal{G}_0$ as in part (3) we can either use that we have $\mathcal{G}$ on Z as in Morphisms, Lemma 07U0 and set $\mathcal{G}_0 = (Z \to Z_0)_*\mathcal{G}$ or we can use Morphisms, Lemma 04CJ and the fact that $\mathrm{Fit}_0(\mathcal{F})$ annihilates $\mathcal{F}$ by More on Algebra, Lemma more-algebra-lemma-fitting-ideal-basics part (6). □